

Azure AI Search vs SQL Server 2025 Hyperscale

Native vector + semantic search comparison · cost crossover model · migration map · East US pricing, May 2026

Your workload

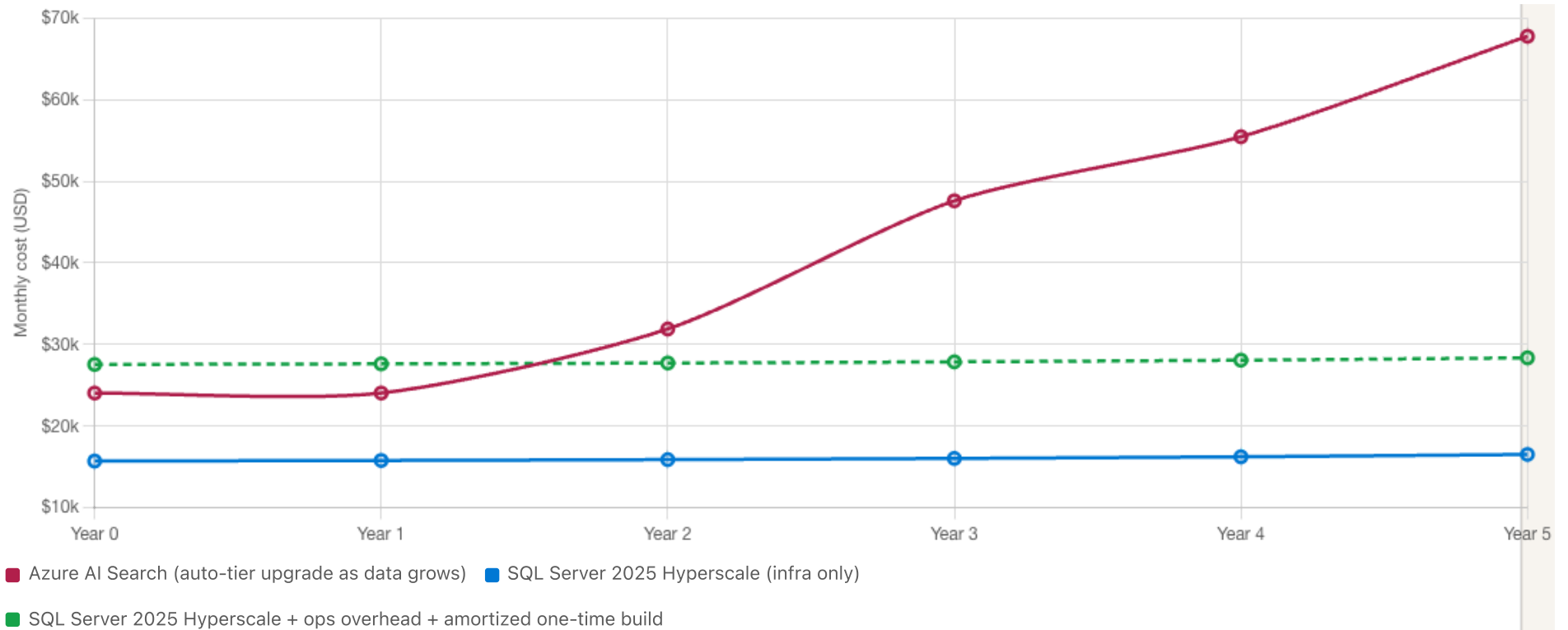
AI SEARCH BASELINE TIER	PARTITIONS (TODAY)	REPLICAS (TODAY)	YOY DATA GROWTH %	TIER STORAGE UTILIZATION %
Standard S2 (512 GB/par✓	6	4	40	60
SEMANTIC RANKER QUERIES / MO (K)	SQL VCORES PER REPLICA	SQL HA REPLICAS (INCL. PRIMARY)	SQL COMPUTE TIER	SQL RESERVATION
500	32	4	Premium Mem-Opt (\$0.2✓	1-year reserved (~35% ✓
SQL OPS OVERHEAD \$/MO (RECURRING)	SQL ONE-TIME BUILD COST \$ (NRE)	PROJECTION HORIZON (YEARS)		
6000	350000	5		

COST CROSSOVER

~ **Year 1.5**

Through year 1.5, **Azure AI Search is cheaper**. After that, SQL Hyperscale + ops overhead crosses below as AI Search is forced into higher tiers.

5-year cost projection



► Assumptions and how the model works

Year-by-year breakdown

YEAR	DATA (TB)	AI SEARCH TIER	PARTITIONS × REPLICAS	AI SEARCH \$/MO	SQL HS \$/MO	SQL HS ALL-IN \$/MO	CHEAPER
Year 0	1.80	Std S2	6 × 4 = 24 SU	\$24,047	\$15,716	\$27,549	AI Search
Year 1	2.52	Std S2	6 × 4 = 24 SU	\$24,047	\$15,789	\$27,623	AI Search
Year 2	3.53	Std S2	8 × 4 = 32 SU	\$31,896	\$15,893	\$27,726	SQL Hyperscale
Year 3	4.94	Std S3	6 × 4 = 24 SU	\$47,594	\$16,037	\$27,871	SQL Hyperscale
Year 4	6.91	Std S3	7 × 4 = 28 SU	\$55,443	\$16,239	\$28,073	SQL Hyperscale

Year 5

9.68

Storage Opt L1

6 × 4 = 24 SU

\$67,759

\$16,523

\$28,356

SQL Hyperscale

Capability parity matrix

CAPABILITY	AZURE AI SEARCH	SQL SERVER 2025 HYPERSCALE	PARITY
Vector type & distance	HNSW + exhaustive KNN, cosine/dot/Euclidean, up to 4 096 dims , scalar & binary quantization, narrow types (int8, float16)	vector(N) data type up to 1 998 dims , VECTOR_DISTANCE function with cosine/dot/Euclidean. <i>Blocks text-embedding-3-large at 3 072 dims.</i>	Partial
ANN vector index	HNSW, managed automatically per index, GA, vector compression baked in	CREATE VECTOR INDEX ... USING DiskANN — still in public preview as of May 2026. Requires PREVIEW_FEATURES = ON. Exact KNN only for < 50K vectors.	Preview
Lexical / BM25 search	Built-in BM25 with analyzers, synonyms, stop-words, language packs	Full-Text Search (CONTAINS/FREETEXT) + Semantic Search (key phrases / similar docs)	Yes
Hybrid (vector + lexical)	Native Reciprocal Rank Fusion (RRF), single query, single response	Hand-rolled CTE combining FTS rank + VECTOR_DISTANCE with manual RRF	DIY
Semantic re-ranker (L2)	Microsoft-hosted cross-encoder, billed per query, captions + answers + query rewrite	Not built-in. Requires self-hosted cross-encoder or AOAI re-rank call	Gap
Integrated vectorization	One-click chunk + embed at index time via AOAI / AI Foundry	sp_invoke_external_rest_endpoint to AOAI, or external pipeline	DIY
Indexers / ingestion connectors	Blob, ADLS, Cosmos, SQL DB, Fabric OneLake, SharePoint, etc. — managed change tracking	ADF, Fabric pipelines, Logic Apps, or custom Functions — manual change tracking	Gap
Document cracking (PDF/DOCX/HTML)	Document Extraction skill + OCR + custom skillsets	Azure AI Document Intelligence called externally; raw text stored in SQL	External

Faceting & filtering	Native facets, filterable fields, scoring profiles	GROUP BY, WHERE, computed/indexed columns, materialized views	Yes
Suggesters / autocomplete	Built-in suggester data structure with fuzzy matching	Trigram or prefix indexes, hand-built	DIY
Geo search	Native Edm.GeographyPoint, geo distance & polygon filters	Native geography / geometry spatial types, spatial indexes	Yes
Agentic retrieval	Native query planner, conversation context, BYOM, billed per reasoning token	Not built-in; orchestrate from app tier with Semantic Kernel / LangChain	Gap
HA / read scale-out	Replicas (1–12), automatic load-balancing	Hyperscale HA replicas (1–4) + named replicas for read scale	Yes
Vertical scale (CPU/RAM)	Fixed per tier — must upgrade tier (S1→S2→S3)	Independent vCore and storage scaling, online vCore changes	Better
Transactional writes	Eventually consistent index updates; no ACID across documents	Full ACID, joins, transactions, foreign keys — same engine as your OLTP	Better
Security: CMK, private link, RBAC	CMK, private endpoints, Entra ID auth, role-based access	TDE w/ CMK, private link, Entra auth, RLS, column encryption	Yes
Backup / PITR	No user-controlled backups (re-index from source)	Continuous PITR, 1–35 day retention, LTR up to 10 years	Better
SLA	99.9% read for ≥2 replicas, 99.9% read/write for ≥3 replicas	99.99% for ≥1 HA replica, 99.995% for zone-redundant HA	Better

Sources: Azure AI Search service limits ([link](#)), Azure SQL Hyperscale docs ([link](#)), SQL Server 2025 vectors ([link](#)).

Can SQL Server 2025 replace Azure AI Search with zero effort?

No. SQL Server 2025 Hyperscale gives you the *primitives* for a retrieval system — the vector data type, DiskANN (preview), Full-Text Search, statistical semantic key phrases. Azure AI Search is a *turnkey retrieval platform*. To get from primitives to platform there is meaningful upfront engineering, ongoing operational overhead, and a small set of capabilities that don't have a SQL equivalent at all.

What does NOT port at all (no SQL equivalent — must approximate or live without)

AI SEARCH FEATURE	CLOSEST SQL SUBSTITUTE	PARITY
Microsoft-tuned semantic re-ranker (L2)	Self-host a cross-encoder (e.g. ms-marco-MiniLM) on Container Apps + GPU, or call AOAI gpt-4o-mini for LLM re-rank. Quality and per-query latency will differ.	Approximate only
Agentic retrieval (autonomous query planner)	Orchestrate from app tier with Semantic Kernel or LangGraph against your SQL hybrid search. Built, not bought.	Build yourself
Integrated vectorization (one-click chunk + embed at index time)	External pipeline calling AOAI embeddings, or sp_invoke_external_rest_endpoint per row. No UI, no managed orchestration.	Build yourself
Indexer connector library (Blob, ADLS, Cosmos, SQL DB, Fabric, SharePoint, ServiceNow...)	ADF / Fabric pipelines / custom Functions. You build and operate each connector and its change-tracking.	Build yourself
4 096-dim vector fields (supports text-embedding-3-large at 3 072)	SQL Server 2025 hard-caps at 1 998 dims. You must use a smaller embedding model (ada-002 at 1 536 fits; 3-large at 3 072 does not).	Hard ceiling
Production-grade ANN vector index (GA)	DiskANN is public preview in May 2026. Exact KNN works but only practical under ~50 K vectors.	Preview risk
Native suggesters / autocomplete with fuzzy match	Trigram or prefix index, hand-built ranker.	DIY

Upfront engineering build-out (one-time, NRE)

WORKSTREAM	ENG-WEEKS (RANGE)	NOTES
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Schema, Hyperscale provisioning, networking, identity	1–2	Straightforward.
Ingestion pipeline (replace indexers)	4–8	One pipeline per source system; change tracking.
Document cracking (Document Intelligence integration)	1–2	If you have rich PDF/DOCX content.
Chunking pipeline	1–2	Recursive splitter or domain-aware.
Embedding generation pipeline	1–2	Batch via Functions / Fabric. Avoid per-row <code>sp_invoke_external_rest_endpoint</code> .
DiskANN vector index setup + tuning	1–2	Preview feature; expect iteration.
Hybrid retrieval w/ RRF in T-SQL	2–4	One stored procedure per query shape; hand-coded RRF.
Semantic re-ranker (host or call AOAI)	4–8	Biggest individual line item. GPU container app + model serving + latency tuning.
Faceting, filters, scoring profiles	2–4	Per query/result shape.
Observability & relevance dashboards	2–3	You lose AI Search query telemetry; rebuild it.
Testing, NDCG/MRR benchmarking, cutover	4–8	Shadow traffic, A/B, parity sign-off.
Total	23–45 eng-weeks	≈ 2 senior engineers for 12–22 calendar weeks.

At a fully-loaded \$8 K / engineer-week (US senior IC, blended), the build cost lands roughly **\$184 K – \$720 K one-time**. The model's default is \$350 K. Add the cost-tab "SQL one-time build cost" input to fold it into the projection as straight-line amortization across your horizon.

15-step build playbook

In order. Where parity is lost, called out inline.

1 Stand up the database

Provision Azure SQL Database Hyperscale (Standard Gen5 or Premium Mem-Opt). Enable Premium-series memory-optimized compute for vector workloads. Configure 1–4 HA replicas to mirror your AI Search replica count and one or more named replicas for read scale-out.

2 Design the schema

Create a `documents` table for source metadata and a `chunks` table with `chunk_id`, `doc_id`, `text nvarchar(max)`, `embedding vector(N)`, plus filterable / facetable columns. Add full-text catalog on the text column.

3 Replace indexers — build an ingestion pipeline

AI Search indexers (Blob, ADLS, Cosmos, SQL, Fabric, SharePoint) become Azure Data Factory pipelines, Fabric Dataflows, or Azure Functions. You own change tracking (high-water mark column or CDC).

4 Replace document cracking

Call Azure AI Document Intelligence for PDF/Office/OCR. Land raw extracted text in `documents.raw_text`. AI Search did this inline; here it's an external skill in the pipeline.

5 Replace the Text Split skill — chunking

Implement chunking in the pipeline (LangChain RecursiveCharacterTextSplitter, Semantic Kernel, or your own UDF). Persist chunks as rows in `chunks`.

6 Replace integrated vectorization — embeddings

Generate embeddings with Azure OpenAI `text-embedding-3-large`. Either call `sp_invoke_external_rest_endpoint` from a stored procedure (per-row, simple) or batch externally in the pipeline (faster, cheaper). Store as `vector(1536)` or `vector(3072)` — note SQL Server 2025 caps at 1 998 dimensions.

7 Build the vector index

`CREATE VECTOR INDEX IX_chunks_emb ON dbo.chunks(embedding) WITH (METRIC = 'cosine', TYPE = 'diskann');` Note: at GA the DiskANN index is single-replica and rebuilt from scratch — plan for batched maintenance windows.

8 Re-implement hybrid retrieval with RRF

Write a single query that runs an FTS lexical search, a vector KNN, ranks each, then combines with hand-coded Reciprocal Rank Fusion. AI Search does this in one line; here it's ~40 lines of CTE/T-SQL per query shape.

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Build a semantic re-ranker — the biggest gap

Choose one: (a) self-host a cross-encoder (e.g. `ms-marco-MiniLM-L-12-v2`) on Azure Container Apps or AKS with GPU and call it from the app tier on the top-50 results; (b) call Azure OpenAI `gpt-4o-mini` to score relevance — much higher quality, much higher cost. Either way: no built-in equivalent to AI Search semantic ranker.

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Recreate scoring profiles and synonyms

Move field-weighting, freshness boosts, and tag boosts into your T-SQL ranking expression. Use FTS thesaurus XML for synonyms; stop-lists for stop words.

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Recreate facets, suggesters, and autocomplete

Facets → `GROUP BY` with indexed columns or pre-aggregated materialized views. Suggesters → prefix or trigram index on the suggester source field; query with `LIKE 'term%'`.

12

Wire up observability & query telemetry

AI Search Insights → use Azure SQL Database Insights, Query Performance Insight, and Application Insights from the app tier. Build a custom relevance dashboard since AI Search's per-query telemetry is not available.

13

Security, network, and identity

Enable TDE with customer-managed key, private endpoints, Entra-only auth, row-level security if you have multi-tenant data, and column-level encryption for PII. All native in Hyperscale.

14

If you used agentic retrieval — rebuild it

AI Search's agentic retrieval (query planning + reasoning tokens) has no SQL equivalent. Build it in the app tier with Semantic Kernel or LangGraph orchestrating multi-turn retrieval against your SQL hybrid search.

15

Cut over with shadow traffic

Run both services in parallel. Replay a representative query log against both and compare top-K results with NDCG / MRR. Only cut over when SQL hybrid + custom re-rank beats AI Search baseline on your relevance benchmark.

Biggest gaps when moving off AI Search: the semantic re-ranker, integrated vectorization, indexer-style connectors, and agentic retrieval. These are

real engineering work — budget them into the cost model on the previous tab via the "SQL ops overhead" input.

Production readiness flag (May 2026): The DiskANN *vector index* in SQL Server 2025 / Azure SQL is still **public preview** — not GA. Exact KNN (no index) is GA but only practical for < ~50K vectors. If your workload needs ANN performance on millions of vectors today, AI Search HNSW is the only GA path on Microsoft's stack. Revisit when DiskANN GAs.

Azure AI Search — per Search Unit (East US, May 2026)

TIER	\$/SU/HR	\$/SU/MO (730 HR)	STORAGE / PARTITION	MAX PARTITIONS	MAX SUS / SERVICE
Basic	\$0.101	\$73.73	15 GB	3	9
Standard S1	\$0.336	\$245.28	160 GB	12	36
Standard S2	\$1.344	\$981.12	512 GB	12	36
Standard S3	\$2.688	\$1,962.24	1 TB	12	36
Storage Opt L1	\$3.839	\$2,802.47	2 TB	12	36
Storage Opt L2	\$7.677	\$5,604.21	4 TB	12	36

SU = partitions × replicas. Service-wide cap is 36 SUs (so with 4 replicas, the maximum partition count is 9 — bumping partitions further requires upgrading the tier). **Azure AI Search does not offer reserved capacity / 1-yr or 3-yr commitment pricing** — all billing is hourly PAYG.

Vector index size cap per partition (separate from data storage cap): Basic 5 GB, S1 35 GB, S2 150 GB, S3 / L2 300 GB, L1 150 GB.

Add-ons

METER	PRICE
Semantic ranker (Standard tier)	\$1.00 / 1 000 queries
Semantic ranker (Free tier)	First 1 000 queries / mo free
Document cracking — image extraction	\$1.00 / 1 000 (volume discounts apply)
Agentic retrieval — medium reasoning	\$0.0001 / 1 000 tokens

Azure SQL Database Hyperscale (East US, May 2026)

COMPUTE SERIES	\$/VCORE/HR (PAYG)	1-YR RESERVED	3-YR RESERVED
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Standard-series Gen5	\$0.18266	~\$0.1187	~\$0.0822
Premium-series	\$0.18266	~\$0.1187	~\$0.0822
Premium-series Memory-Optimized	\$0.25572	~\$0.1663	~\$0.1150
DC-series (confidential)	\$0.36500	\$0.2374	\$0.1644

Each HA replica and named replica is billed at the same compute rate as the primary. Reserved capacity discounts are approximate (~35% for 1-yr, ~55% for 3-yr).

Storage & backup

METER	PRICE
Hyperscale data storage (Single DB)	\$0.10 / GB / month
Hyperscale data storage (Elastic Pool)	\$0.25 / GB / month
Hyperscale I/O	\$0.20 / 1 M I/O ops
Backup storage (LRS / ZRS short-term)	Included up to 100% of DB size
LTR backup (LRS / ZRS)	\$0.025 / GB / month
LTR backup (RA-GRS / RA-GZRS)	\$0.05 / GB / month

Source: Azure Retail Prices API (prices.azure.com/api/retail/prices), East US region, retrieved May 20, 2026.